



Anirudh Prahlad Joshi

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Professional Summary

Data Science Master's student at TU Hamburg with hands-on experience in software engineering, machine learning, and scalable data systems within automotive and industrial environments. Skilled in Python, PyTorch, SQL, and data engineering workflows, with applied experience in computer vision, NLP, and real-time analytics. Experienced working in cross-functional and agile teams, with a focus on developing reliable, high-performance solutions that translate data into real-world impact.

Authorized to work in the European Union with a valid EU work permit; no visa sponsorship required.

Technical Skills

Programming Languages: Python, C++, C

Machine learning libraries & AI: PyTorch, scikit-learn, NLP, Computer Vision, Generative AI, RAG, Langchain, LangGraph, LoRA

Data Engineering & Distributed Systems: Apache Kafka, Apache Flink, PySpark

Cloud & DevOps: Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, CI/CD

Databases: SQL, MongoDB

Tools & Frameworks: Git, Vue.js

Projects

Federated Sentiment Analysis

Apr 2025 – Jul 2025

- Designed a low-latency streaming pipeline using Apache Kafka and Apache Flink for large-scale data ingestion and processing.
- Implemented continuous model retraining using streaming data, enabling adaptive and real-time ML predictions.
- Deployed the system on Google Cloud Platform using Docker, ensuring scalability and fault tolerance.
- Demonstrated understanding of batch vs streaming trade-offs and real-time inference systems.

Technologies: Apache Kafka, Apache Flink, JavaScript, Docker, Google Cloud Platform (GCP)

Understanding V in Multi-Modal Language Models

Feb 2025 – Sep 2025

- Investigated limitations in visual understanding of multimodal language models for VQA tasks by designing controlled experiments using custom datasets, enabling systematic evaluation of model behavior and weaknesses.

- Built a custom dataset from COCO images and GPT-4.1-generated question-answer pairs, increasing training data diversity and improving evaluation coverage.
- Trained and optimized BERT-BEiT-based vision-language models in PyTorch, achieving improved accuracy and more stable performance across VQA benchmarks.

Technologies: PyTorch, Python, GPT-4.1, Computer Vision

Debiasing TextVQA

Jan 2022 – Jul 2022

- Addressed dataset bias in TextVQA models, integrated VQA dataset as an external dataset and normalization techniques, improving fairness and generalization performance.
- Evaluated model robustness across diverse benchmarks, strengthening research validity.

Technologies: PyTorch, Python, NumPy, Computer Vision

Work Experience

Student Assistant — Institute of Production Management and Technology (TUHH)

Feb 2025 – Present

- Designed and implemented real-time data pipelines using Python, enabling scalable data processing and improved visibility into production workflows.
- Developed a data-driven web application to digitalize production processes, improving efficiency and decision-making for stakeholders.
- Collaborated in a cross-functional agile environment with backend developers and domain experts to translate business requirements into technical solutions.
- Built interactive dashboards and monitoring tools, enhancing accessibility and usability of operational data.
- Contributed to system design decisions focusing on scalability, reliability, and maintainability.

Technologies: Python, MongoDB, Docker, Vue.js, Git

Associate Software Engineer — Bosch Global Software Technologies

Jul 2023 – Sep 2024

- Developed and optimized data-intensive components for automotive infotainment systems, improving query performance and reducing latency.
- Implemented efficient data indexing and search algorithms, enhancing system responsiveness and user experience.
- Identified and resolved performance bottlenecks and data inconsistencies, improving overall system reliability.
- Worked in a cross-functional agile team, collaborating with software, hardware, and UX teams to deliver production-ready features.
- Conducted integration, regression, and system-level testing on embedded automotive systems.

Technologies: C++, SQLite, Qt, Git

Education

Hamburg University of Technology (TUHH)

Master's in Data Science

Oct 2024 – March 2027 (Expected)

K.L.E Technological University

Bachelor's in Computer Science and Engineering

Aug 2019 – Jun 2023

Languages

English (C1) — German (A1) — Kannada (Native)